
Backbone-Modular Neuro-Symbolic Post-Correction for OCT Denoising via Explicit Predicate Maps and Stable Fuzzy Allocation

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ABSTRACT Optical coherence tomography is read through the contrast between retinal layers and the sharpness of their boundaries. A denoiser is trained on a pixel error instead, so it can raise fidelity and still return a scan that has lost them. Nothing sits between the denoiser and the reader to say what a good scan looks like and repair where one falls short. We make that repair a problem of its own, solved after denoising with the denoiser frozen. Four things are new. Six clinical properties in closed form read only the noisy input and the denoised output, each returning a score and a map of where it fails, so they need no clean image and run on any scanner. Five rules in fuzzy logic turn those maps into a per-pixel allocation, whose response to a shift in evidence we bound by an exact, attained constant. A bounded proposal is admitted according to how many of three constraints hold, we prove when the correction cannot lower contrast, and every change traces to a named property at a named place. One set of rules, budgets and weights serves four denoisers of different families, only a small head refitted. All six measures rise on two of the four, five of six on the others, contrast by 3.5 to 11.3 percent, speckle suppression by 23.7 to 57.1 and signal to noise by 8.0 to 23.0, at 0.17 to 0.98 dB of fidelity. Carried unchanged to two other datasets, 40 of 48 improve.

INDEX TERMS Clinical quality assessment, fuzzy logic, image denoising, neuro-symbolic reasoning, optical coherence tomography.

I. Introduction

OPTICAL coherence tomography (OCT) is the standard non invasive way to image the retina [1]. Every OCT image carries speckle, a multiplicative disturbance that arises from coherent light scattering inside tissue [2]. Speckle blurs the border between retinal layers, lowers the contrast between tissue and background, and makes automatic layer segmentation less reliable [3].

Deep denoisers remove much of that disturbance, but they are almost always trained to minimize the squared error against a clean reference, so a denoiser can raise the peak signal to noise ratio (PSNR) of a scan while making it less useful to a clinician. Layer boundaries lose definition, the contrast between tissue and the dark vitreous falls, and thin structures fade into the background. None of these losses is visible in a PSNR number.

A. Motivation

The gap we address is not a missing denoiser. It is that nothing sits between the denoiser and the clinical reader to state what a good OCT image should look like and to repair the ways a denoised one falls short. Another neural block would hide the reason for every change inside learned weights, while a clinician, a regulator or a reviewer who sees a modified scan needs to know which property was judged deficient, where, and how large a change was allowed. We therefore treat post-correction as its own problem, keeping the denoiser frozen and attaching a small wrapper written as named image properties, explicit rules over them and explicit limits on the size of any change. Every learned map in the decision path is bounded, and its effect reaches the image only through a named rule.

B. New ideas in this paper

The new ideas are four. Edge filters, gating maps and fuzzy operators are not among them.

- 1) Six clinical properties that need no clean image. Section IV-D defines six properties computed from the noisy input and the denoiser output alone, so they are available on any scanner at test time. Each returns a score for the whole image and a map of where it fails, both as formulas.
- 2) A rule layer whose decision cannot jump. Section IV-E builds the allocation from five rules joined by a conjunction with bounded slope, Section V proves that a shift in the evidence moves the decision by at most a constant that the rule layer itself fixes, and Section VII-G measures the response the images actually produce against it.
- 3) A safety layer that decides how much of a proposal to keep. Section IV-G states three constraints at three scales, and how many of them hold sets the fraction of the proposed change that is applied. A margin condition under which the correction cannot lower the contrast to noise ratio is proved and then tested on every test image.
- 4) One design for four denoisers. Only the head that reads backbone features changes with the backbone. The rules, the budgets, the thresholds and the loss weights do not, the shared wrapper is a small fraction of the denoiser it is attached to, and Section VII-C carries it to two other datasets with no adaptation at all.

II. Related Work

A. Denoising of OCT images

Learned denoisers for OCT fall into three families. Convolutional networks such as DnCNN [4] learn the noise as a residual and subtract it. Designs that gate or reweight channels rather than pass them through a nonlinearity, such as NAFNet [5] and KBNNet [8], control what each contributes. Transformers such as SwinIR [6] use windowed self attention. Methods built for OCT address the scarcity of paired data directly, through triplet cross fusion [9] and sparse representation [11], although the earlier of those still needs one high quality scan of the same eye [10].

All of these are trained to reduce a pixel error, so the quantity they optimize is not the quantity a clinician reads. Work on quality assessment for OCT [12] measures clinical quality but does not act on it. This paper closes that loop, turning measured quality into a correction applied under stated limits.

B. Confidence and uncertainty in medical imaging

Deep ensembles and their relatives [13] are the standard way to attach a confidence to a prediction. In OCT the networks that segment layers [3] or judge scan qual-

ity [12] report an accuracy rather than an uncertainty, so a reader learns how often the model is right and never where it is unsure, and the value is reported and then left alone. We feed such a map into the correction decision instead, and Section VII-F reports that the map we built is not a confidence in this sense.

C. Symbolic reasoning and expert networks joined with neural networks

Neuro symbolic computing [14] couples a learned model with explicit rules, and differentiable fuzzy logic makes the coupling trainable because the connectives become smooth functions of continuous truth values [7]. In medical imaging the pattern has generated spinal reports by pairing a generative network with a symbolic graph of domain knowledge [20] and predicted patient deterioration with a probabilistic logic program over convolutional features [21]. Fuzzy membership functions have been embedded in networks that process OCT directly, where a rough fuzzy discretization guides the segmentation of fluid lesions [22], and pixel level fuzzy rules have replaced learned feature extraction, so what the model attends to reads as a logical statement [23].

A parallel line builds an image model from cooperating experts. Steered mixtures of experts combine constant experts through soft gating kernels with an explicit steering term [24], also for OCT speckle removal [25], and recent restoration networks route an input through experts of different capacity according to its difficulty [26]. Outside imaging, evolving fuzzy models identify actuator dynamics [17], learned models are joined to a physical forward operator in ultrasound tomography [18], and sensitivity to initial conditions is treated as a source of uncertainty in simulation [19]. All rest on the principle we rely on, that a learned component is easier to trust when part of the decision is written in a language a person can read. What is missing is a rule layer that acts on the output of a strong denoiser, names the clinical properties it protects, and limits how far it may move the image, which is the object studied here.

III. Notation and Problem Statement

A. Notation

Images are real valued arrays of height H and width W with values in $[0, 1]$. We write $\mathbf{y}$ for the noisy B-scan that comes off the scanner and $\mathbf{x}$ for the clean reference obtained by averaging many repeated scans of the same location. A pixel index is written p , so y_p is one pixel of $\mathbf{y}$.

The symbol $\odot$ is the Hadamard product, $(\mathbf{t} \odot \mathbf{t}')_p = t_p t'_p$. Table 1 lists the symbols that recur across sections, and a symbol used in one place only is defined where it is introduced.

TABLE 1. Symbols used in the paper, in order of first use.

Symbol	Meaning
$\mathbf{y}$	Noisy input B-scan
$\mathbf{x}$	Clean reference B-scan
f_θ	Frozen denoising backbone with parameters θ
$\mathbf{b}$	Backbone output, $\mathbf{b} = f_\theta(\mathbf{y})$
ϕ	Trainable parameters of the wrapper
$\mathbf{c}$	Cooperation map with values in $[0, 1]$
$\mathbf{u}$	Demand map, $\mathbf{u} = \mathbf{1} - \mathbf{c}$
s_i	Score of clinical property i , a number in $[0, 1]$
$\mathbf{m}_i$	Failure map of clinical property i , values in $[0, 1]$
τ_i	Decision threshold for property i
Δ	Raw multiplicative gain proposed by the gain network
λ	Per pixel strength map with values in $[0, 1]$
$\mathbf{a}$	Allocation map produced by the rule layer
α	Gain after allocation, strength and clamping
$\mathbf{g}$	Background gate that suppresses change in the background
$\mathbf{v}$	Gated multiplicative change added to the backbone output
δ_e	Additive edge correction
$\mathbf{q}$	Candidate image before the safety decision
$\hat{\mathbf{x}}$	Final output of the wrapper
$\mathcal{T}, \mathcal{B}$	Sets of tissue pixels and background pixels
$\mu_{\mathcal{T}}, \mu_{\mathcal{B}}$	Mean value over tissue and over background
$\sigma_{\mathcal{B}}$	Standard deviation over background

B. The optimization problem

The denoiser is given and is not changed. The task is to produce an image that beats $\mathbf{b} = f_\theta(\mathbf{y})$ on named clinical measures, stays close to it everywhere, and can explain every change it makes. The only free quantities are the wrapper parameters ϕ , namely the cooperation head, the gain network, the edge network, the strength predictor and the rule layer, while the backbone parameters θ are held fixed throughout. Let $\mathcal{D}$ be a set of paired scans and $\hat{\mathbf{x}}(\phi)$ the output of the wrapper. We minimize the expected value of a weighted sum of six terms with fixed weights w_c, w_p, w_o, w_s and w_r ,

$$\min_{\phi} \mathbb{E}_{(\mathbf{y}, \mathbf{x}) \sim \mathcal{D}} \left[\mathcal{L}_{\text{fid}} + w_c \mathcal{L}_{\text{clin}} + w_p \mathcal{L}_{\text{pred}} + w_o \mathcal{L}_{\text{coop}} + w_s \mathcal{L}_{\text{edge}} + w_r \mathcal{L}_{\text{reg}} \right], \quad (1)$$

where $\mathcal{L}_{\text{fid}}$ keeps the output near the clean reference, $\mathcal{L}_{\text{clin}}$ rewards improvement of the clinical measures over the backbone, $\mathcal{L}_{\text{pred}}$ asks four property scores to rise above what the backbone already reaches, $\mathcal{L}_{\text{coop}}$ ties the amount of correction to the estimated demand, $\mathcal{L}_{\text{edge}}$ supervises the gradient field, and $\mathcal{L}_{\text{reg}}$ collects the bounds on how large and how rough the change may be together with four terms that protect the background. Section VI gives every term with its weight.

Three constraints are checked on every image and stated formally in Section IV-G. The aggregate clinical quality should not fall by more than a fixed tolerance, no single property should fall by more than a fixed tolerance while another rises, and the relative change from the backbone output should stay below a fixed bound. How many hold decides how much of the change is kept, so the rule is a majority test and not three hard guarantees. Every logical connective is a piecewise linear surrogate, so the objective is differentiable in ϕ including the rule

layer and is solved with AdamW [16]. The constraints enter training as penalties and are evaluated exactly on each image at test time.

IV. Method

A. Overview

Fig. 1 shows the method as six stages, taken in order in the subsections that follow. Stage one asks where the backbone output can be left alone, two what change is available, three what is wrong and where, four where it should be applied, five how, and six whether it is allowed at all.

B. Stage one, the backbone and the cooperation map

The backbone is frozen after training and gives $\mathbf{b} = f_\theta(\mathbf{y})$. The cooperation map $\mathbf{c}$ says where $\mathbf{b}$ can be left alone and where a change is wanted. A two layer network h_ϕ reads a feature tensor $\mathbf{z}$ from inside the backbone and produces it,

$$\mathbf{c} = \sigma \left(\frac{h_\phi(\mathbf{z}) + \beta_0}{\tau_0} \right), \quad (2)$$

where σ is the logistic function, $\tau_0 > 0$ is a learnable temperature and β_0 a learnable offset. The complementary demand map is $\mathbf{u} = \mathbf{1} - \mathbf{c}$.

Only this head depends on the backbone, since architectures expose feature tensors of different width. We read the first encoder stage of NAFNet and KBNet, the first convolution of DnCNN and the shallow feature convolution of SwinIR, and the head is a pair of convolutions sized to whichever tensor it is given, the widths listed in the supplementary file. Everything after this stage is shared, which is the precise sense in which the method is modular rather than architecture free.

C. Stage two, the two correctors

Two changes of deliberately different kinds are proposed.

The gain network proposes a multiplicative change. It runs at quarter resolution and emits a brightness channel through a rectifier, so it can only brighten, and a contrast channel through a hyperbolic tangent, so it can go either way. Their sum is scaled by $\alpha_{\text{max}} = 0.15$ to give the raw gain Δ . A multiplicative form suits speckle, which is itself multiplicative [2], so bright tissue receives a larger absolute change than dark background for the same relative adjustment.

The edge network proposes an additive change δ_e from the intensity normalized residual, the edge magnitude of the backbone output, that output itself and an unsharp mask. It is bounded to ± 0.08 and scaled by a learned factor. An additive form suits boundaries, which should be sharpened by the same amount in a bright layer as in a dark one.

A third small network predicts a per-pixel strength map λ with values in $[0, 1]$ from the failure maps of

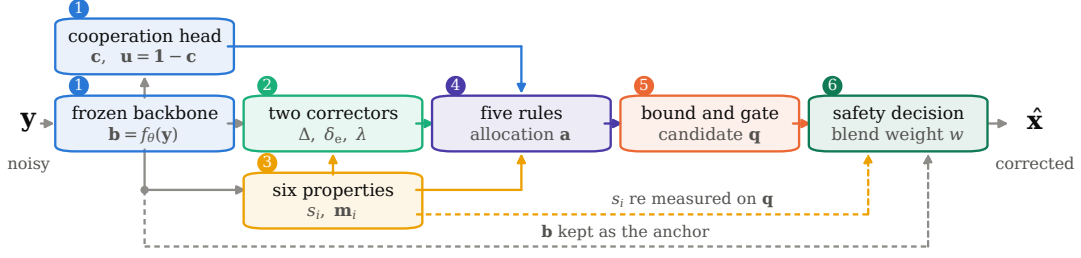


FIGURE 1. The six stages. A frozen backbone and a small head produce the image and the cooperation map. Two networks propose changes. Six properties are measured. Five rules allocate. The change is bounded and gated. Three constraints decide how much is kept.

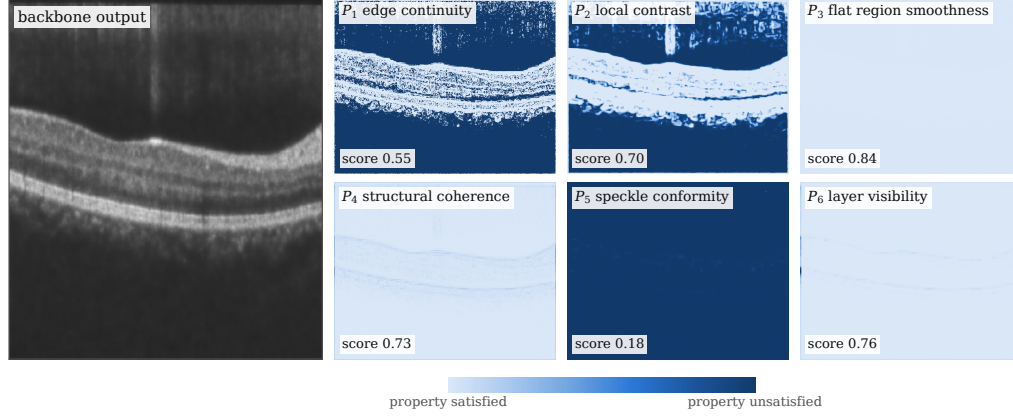


FIGURE 2. The six failure maps on one backbone output, with the score each property returns for the whole image. Dark means strong local evidence that the property is unsatisfied. The maps disagree with each other, which is the point of keeping six of them. Edge continuity and local contrast fire along the layer boundaries and through the textured tissue between them, speckle conformity fires almost everywhere on this image and drives its low score, and the remaining three are close to satisfied and stay pale. The allocation of Equation (4) is built from these maps, so a reader can see which property asked for a change at any pixel.

Stage three. It modulates how much of the raw gain survives at each pixel.

D. Stage three, the six clinical properties

Each of the six properties names a way a denoised OCT image is known to fail and each is written as a formula so it can be recomputed by anyone. Table 2 gives, for every property, the reason it exists, its score $s_i \in [0, 1]$ as a weighted sum of named subterms, its map $\mathbf{m}_i \in [0, 1]^{H \times W}$ in closed form, where that map is large, and the threshold τ_i that separates satisfied from unsatisfied, with every subterm defined beneath the table. A larger score means the property is better satisfied and a larger map value means stronger local evidence that it is not, and Fig. 2 shows all six maps on one image. None of these quantities needs a clean reference. They are computed from the noisy input and the backbone output only, so they remain available at test time on a scanner the method has never seen.

E. Stage four, the rule layer

The rule layer decides where the gain proposal should act through five named rules, each casting a signed vote.

Each property score is first turned into a soft failure signal by comparing it with its threshold,

$$f_i = \sigma(\kappa(\tau_i - s_i)), \quad \kappa = 10, \quad (3)$$

so f_i is near one when property i is failing. The same soft comparison tests the cooperation map and the strength of a proposal against their own thresholds. Each corrector produces that strength as a per-pixel map from the backbone output and the failure maps it serves, large where its property is far below threshold, and it is multiplied by one half plus the demand map before the rules fire, so the same proposal looks stronger where the head reports low cooperation. Conjunction uses the Lukasiewicz form $\text{AND}(a, b) = \max(0, a + b - 1)$, piecewise linear with a bounded slope, which is what makes the bound of Section V tight.

The five rules follow. Rule one fires when the cooperation map is high and the proposal weak, and votes against changing the image. Rule two fires when the map is low and the proposal strong, and votes for. Rule three fires when a relevant property is failing, and votes for. Rule four fires when map and strength are both middling, and votes weakly for. Rule five fires when several correctors want the same pixel, and votes against. Write r_1 to r_5 for the truth values.

The allocation at each pixel is the bounded weighted vote

$$a = \underbrace{B_{\min} + (B_{\max} - B_{\min}) \sigma(\beta)}_{\text{base, no rule firing}} - C_1 w_1 r_1 + C_2 w_2 r_2 + C_3 w_3 r_3 + C_4 w_4 r_4 - C_5 w_5 r_5, \quad (4)$$

where β is a learnable base logit, $w_k = \sigma(\theta_k) \in (0, 1)$ are learnable rule weights and the budgets are $B_{\min} = 0.25$, $B_{\max} = 0.50$, $C_1 = 0.15$, $C_2 = 0.25$, $C_3 = 0.15$, $C_4 = 0.10$ and $C_5 = 0.10$.

The budgets satisfy $B_{\max} + C_2 + C_3 + C_4 = 1$ and $B_{\min} - C_1 - C_5 = 0$. Since every truth value lies in $[0, 1]$ and every weight in $(0, 1)$, Equation (4) lies in $[0, 1]$ for every input and every parameter value, and no clipping is ever needed. Without the identity a clamp is needed, the allocation saturates at one end of it, as it did on every pixel of the submitted version of this work, the rule weights receive no gradient, and the stability result of Section V is vacuous.

F. Stage five, applying the change

The admitted gain is the raw proposal modulated by the allocation and the strength map, then bounded,

$$\alpha = \text{clamp}(\Delta \odot \mathbf{a} \odot \lambda, -0.20, 0.20). \quad (5)$$

A gate suppresses any change in the dark background,

$$\mathbf{g} = \sigma(20(\mathbf{b} - q_{0.15})), \quad (6)$$

where $q_{0.15}$ is the fifteenth percentile of the backbone output of that image. A percentile rather than a fixed level makes the gate work across scanners of different brightness. The second form, used where the selection of Section VI-C chose it, is a soft step at the per image Otsu threshold followed by a maximum filter of fixed radius, so the gate is identically zero beyond that radius from tissue.

The gated gain is applied multiplicatively to give the change $\mathbf{v}$, with the brightening part smoothed before it is added so that it cannot create a new high frequency edge,

$$\mathbf{v} = (\text{smooth}(\text{ReLU}(\alpha) \odot \mathbf{b}) + (\alpha - \text{ReLU}(\alpha)) \odot \mathbf{b}) \odot \mathbf{g}, \quad (7)$$

where smooth is a three by three mean filter, so a brightening is spread over a neighborhood while a darkening acts where it was proposed. In the background, the complement of the gate eroded by two pixels, a masked mean filter of the same size is blended in at half strength, which lowers the background standard deviation without letting blur cross into tissue. The edge proposal is then added and the result clipped to the valid range,

$$\mathbf{q} = \text{clamp}(\mathbf{b} + \mathbf{v} + \delta_e \odot \mathbf{g}, 0, 1). \quad (8)$$

G. Stage six, the safety decision

The candidate $\mathbf{q}$ is not returned directly. Three constraints are checked on it.

Aggregate quality must not fall. Define for any image $\mathbf{t}$ the energy

$$E(\mathbf{t}) = 1 - \frac{1}{6} \sum_{i=1}^6 s_i(\mathbf{t}) + 0.1 \sum_{i=1}^6 \max(0, 0.5 - s_i(\mathbf{t})), \quad (9)$$

which is small when the image is good and whose second term penalizes any property below one half, so a good average cannot hide one broken property. The constraint is $E(\mathbf{q}) \leq E(\mathbf{b}) + 0.10$.

No property may be sacrificed for another. The second constraint requires that at least one property improves by more than 0.01 and that no property falls by more than 0.10.

The image must not move far. The third constraint requires $\|\mathbf{q} - \mathbf{b}\|_2 / \|\mathbf{b}\|_2 < 0.2$.

Let $n \in \{0, 1, 2, 3\}$ be the number of constraints that hold. The output is

$$\hat{\mathbf{x}} = (1 - w) \mathbf{b} + w \mathbf{q}, \quad w = \begin{cases} 1, & n \geq 2, \\ 0.05 + 0.10 \frac{n}{3}, & n < 2. \end{cases} \quad (10)$$

The candidate is accepted in full whenever any two constraints hold, so one violation does not by itself withhold the change. Below two the output falls back to the backbone while keeping five percent of the change when none holds and about eight when one does, so the decision stays visible rather than being silently discarded. This is a majority test on three measured conditions and not an enforcement of three bounds, and Section VII-H reports how often each holds.

V. Stability and Guarantees

Section VII-G tests both results of this section on real images.

A. What stability means here

This is not stability in the sense of Lyapunov and we claim no fixed point. The quantity that must not behave erratically is the decision of the rule layer, a function of the five rule truth values, which are exactly what shift when the backbone or the scanner changes.

Definition 1 (Stability of the allocation). Let $\mathbf{r} = (r_1, \dots, r_5) \in [0, 1]^5$ be the vector of rule truth values at one pixel and let $a(\mathbf{r})$ be the allocation of Equation (4) at that pixel with the parameters held fixed. The allocation is said to be stable with constant L if

$$|a(\mathbf{r}) - a(\mathbf{r}')| \leq L \|\mathbf{r} - \mathbf{r}'\|_\infty \quad \text{for all } \mathbf{r}, \mathbf{r}' \in [0, 1]^5. \quad (11)$$

In words, L is the worst case ratio between a change in the decision and the change in the evidence that caused

TABLE 2. The six clinical properties in closed form. Here e is the Sobel edge magnitude of $\mathbf{b}$, $\bar{e}$ its mean, $b_e = \mathbf{1}[e > \bar{e}]$, σ_k and Var_k the local standard deviation and variance over a $k \times k$ window, $\bar{\sigma}$ the mean of σ_9 , $v(\cdot)$ the depth profile averaged across width, ∇_z the derivative along depth, ρ_h the one pixel horizontal autocorrelation, sigm the logistic function, ε a small constant, and cv the local coefficient of variation of the log residual. The named subterms are expanded in the supplementary file.

P_i	Property	Score s_i	Map $\mathbf{m}_i$	Where the map is large	τ_i
P_1	edge continuity	$0.4 c_{\text{cont}} + 0.3 c_{\text{efr}} + 0.3 c_{\text{ang}}$	$(1 - \bar{e})(1 - b_e)$, $\bar{e} = e/(\max e + \varepsilon)$	off detected edges, where a boundary should continue but does not	0.60
P_2	local contrast	$0.20 s_{\text{range}} + 0.35 s_{\text{cnr}} + 0.20 s_{\text{var}} + 0.25 s_{\text{cov}}$	$[\text{ReLU}(\bar{\sigma} - \sigma_9(\mathbf{b})) / (\bar{\sigma} + \varepsilon) + 0.3 \mathbf{1}[\sigma_9(\mathbf{b}) < 0.01]]_{[0,1]}$	where σ_9 falls below the image mean, and in nearly flat tissue	0.50
P_3	flat smoothness	$0.6 q_{\text{var}} + 0.4 q_{\text{res}}$	$F \odot \sigma_5(\mathbf{b}) / (\max \sigma_5(\mathbf{b}) + \varepsilon)$	inside flat regions that still carry residual variation	0.75
P_4	structural coherence	$0.4 s_{\text{self}} + 0.3 s_{\text{reg}} + 0.3 s_{\text{vert}}$	$e(e) / (\max e(e) + \varepsilon)$	where the edge map is itself rough, a sign of broken structure	0.50
P_5	speckle conformity	$0.7 \text{mean}(\exp(- cv - cv^* /0.15)) + 0.3 \exp(-5 \rho_h(z))$	$1 - \exp(- cv - cv^* /0.15)$	where cv departs from the value expected for OCT speckle	0.30
P_6	layer visibility	$0.4 s_{\text{clr}} + 0.3 s_{\text{dep}} + 0.3 s_{\text{smo}}$	$\text{Var}_7(e) / (\max \text{Var}_7(e) + \varepsilon)$	where the local variance of the edge map is high across depth	0.70

it. Without such a bound the layer could swing from no correction to all of it because one score moved slightly, which is what a clinical user cannot audit.

B. The rule layer is stable, with an exact constant Theorem 1 (Allocation stability). The allocation of Equation (4) satisfies Definition 1 with

$$L = C_1 w_1 + C_2 w_2 + C_3 w_3 + C_4 w_4 + C_5 w_5, \quad (12)$$

where $w_k = \sigma(\theta_k)$ are the learned rule weights. The constant is attained, so it is the exact Lipschitz constant and not merely an upper bound. Moreover $L < C_1 + C_2 + C_3 + C_4 + C_5 = 0.75$ for every possible setting of the parameters.

Proof:

The allocation is affine in $\mathbf{r}$ with coefficients $\pm C_k w_k$, so subtracting two values and applying the triangle inequality gives $|a(\mathbf{r}) - a(\mathbf{r}')| \leq \sum_k C_k w_k |r_k - r'_k| \leq L \|\mathbf{r} - \mathbf{r}'\|_\infty$. The bound is attained when each $r_k - r'_k$ takes the sign of its coefficient, and $L < 0.75$ because every C_k is positive and every w_k lies strictly inside $(0, 1)$. ■

Corollary 1 (Behavior under a change of backbone). If replacing one backbone by another changes the rule truth values at a pixel by at most ϵ in the maximum norm, the parameters unchanged, the allocation there changes by at most $L\epsilon$.

This concerns the routing rule alone, not whether image quality transfers, which Section VII-C settles by measurement.

C. When the correction cannot reduce contrast

Write $\text{CNR}(\mathbf{t}) = (\mu_{\mathcal{T}}(\mathbf{t}) - \mu_{\mathcal{B}}(\mathbf{t})) / \sigma_{\mathcal{B}}(\mathbf{t})$ for the separation between tissue and background divided by the variability of the background.

Theorem 2 (Contrast margin). Let $\eta_{\mathcal{T}} = \mu_{\mathcal{T}}(\mathbf{q}) - \mu_{\mathcal{T}}(\mathbf{b}) \geq 0$ be the gain in tissue mean and let $\eta_{\mathcal{B}} \geq \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b})$ bound the leakage into the background. Suppose the tissue and background sets are the same for $\mathbf{b}$, $\mathbf{q}$ and $\hat{\mathbf{x}}$, that $\sigma_{\mathcal{B}}(\mathbf{b}) > 0$ and the correction does not

raise the background variability, so $\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$, and that the bounded numerator $(\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}})$ is non negative, which holds whenever the backbone already separates tissue from background and $\eta_{\mathcal{T}} \geq \eta_{\mathcal{B}}$. Then the output of Equation (10) satisfies

$$\text{CNR}(\hat{\mathbf{x}}) \geq \frac{(\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}})}{\sigma_{\mathcal{B}}(\mathbf{b})}. \quad (13)$$

In particular, whenever $\eta_{\mathcal{T}} > \eta_{\mathcal{B}}$ the correction cannot reduce the contrast to noise ratio.

Proof:

The mean over a fixed pixel set is linear, so $\mu_{\mathcal{T}}(\hat{\mathbf{x}}) = \mu_{\mathcal{T}}(\mathbf{b}) + w\eta_{\mathcal{T}}$ and $\mu_{\mathcal{B}}(\hat{\mathbf{x}}) \leq \mu_{\mathcal{B}}(\mathbf{b}) + w\eta_{\mathcal{B}}$, which bounds the numerator of $\text{CNR}(\hat{\mathbf{x}})$ below by the numerator of Equation (13). On the background $\hat{\mathbf{x}}$ is the same convex combination pixel by pixel, so Minkowski's inequality on the centered values gives $\sigma_{\mathcal{B}}(\hat{\mathbf{x}}) \leq (1 - w)\sigma_{\mathcal{B}}(\mathbf{b}) + w\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$. Dividing the two bounds gives the claim, and $w > 0$ in every branch of Equation (10). ■

Three parts of the design make these hypotheses hold in practice. The gate of Equation (6) keeps $\eta_{\mathcal{B}}$ small by suppressing change in the dark region, the brightness branch cannot be negative, which keeps $\eta_{\mathcal{T}}$ non negative, and the background smoothing lowers $\sigma_{\mathcal{B}}(\mathbf{q})$ rather than raising it.

Both results are deliberately narrow. The first bounds how far the routing decision can move, not whether it is correct. The second concerns contrast alone. Neither promises that the output is closer to the clean reference, which Section VII-B shows it usually is not. The guarantees concern the wrapper, not the denoiser underneath it.

VI. Training

A. The six loss terms

Let $\mathbf{x}$ be the clean reference, $\mathbf{b}$ the backbone output and $\hat{\mathbf{x}}$ the wrapper output. The objective of Equation (1) has the following terms.

Fidelity. Let $\Delta = \text{PSNR}(\mathbf{b}) - \text{PSNR}(\hat{\mathbf{x}})$ be the drop against the backbone in decibels and Δ_s the corresponding drop in the structural similarity index [15]. Neither

is charged until it exceeds a dead zone,

$$\mathcal{L}_{\text{fid}} = 5 [\Delta - d]_+ + 15 [\Delta - d - 0.4]_+^2 + 20 [\Delta_s - 0.005]_+, \quad (14)$$

where $[\cdot]_+$ is the positive part. The term is flat while the drop stays below d decibels, linear for the next four tenths and quadratic beyond, so a small loss of fidelity is free and a large one expensive. The dead zone d is one of the two quantities selected per backbone in Section VI-C.

Clinical improvement. This term charges the wrapper only when a measure falls short of a target taken from the backbone, so a gain in one measure cannot be bought by a collapse in another. Let q_m be the value of quantity m on the wrapper output and $t_m = \kappa_m q_m(\mathbf{b})$ the target, the backbone value of the same quantity scaled by a fixed factor. Writing $\varepsilon_m = t_m - q_m$,

$$\mathcal{L}_{\text{clin}} = \frac{1}{\sum_m w_m} \sum_{m \in M} w_m \left(3[\varepsilon_m]_+ + [\varepsilon_m]_+^2 + \frac{1}{2} s_m \varepsilon_m^2 \right). \quad (15)$$

The set M holds nine quantities, the three preservation scores of local contrast, boundary gradient and texture, then EPI, CNR, BS, ENL, SNR and the ratio of the backbone squared error to the wrapper squared error. Their weights w_m are 1, 5, 1, 15, 40, 5, 8, 40 and 10 and their factors κ_m are 1.12, 1.12, 1.00, 1.05, 1.20, 1.12, 1.10, 1.15 and 0.87, the last allowing about six tenths of a decibel. The indicator s_m is one for ENL, SNR and the squared error ratio, a symmetric pull that holds those three near target. None of the nine is used at test time.

Preservation scores. This term acts on four scores g_j separate from the six properties of Table 2, comparing corrected against backbone on local contrast, the gradient across layer boundaries, the local variance carrying texture and the gradient magnitude at edges. With $\delta_j = g_j(\hat{\mathbf{x}}) - g_j(\mathbf{b})$, the backbone value fixed,

$$\mathcal{L}_{\text{pred}} = \sum_{j=1}^4 v_j \left(3[-\delta_j]_+ - 2.5[\delta_j]_+ + 2 \left[\frac{1}{2} - g_j(\hat{\mathbf{x}}) \right]_+ \right), \quad (16)$$

where v_j is 3, 2, 2 and 1 in that order. A drop costs more than a rise earns, and any score below one half is charged again. The thresholds τ_i play no part here. They enter Equation (3), but the rule layer that evaluates it runs without gradient tracking, so no path reaches them and all six hold their Table 2 values throughout training.

Cooperation. The wrapper should act where the backbone leaves room and stay quiet elsewhere. We take the Pearson correlation ρ between the demand map $\mathbf{u}$ and $\boldsymbol{\pi}$, the summed proposal strength the rule layer reads, and penalize disagreement,

$$\mathcal{L}_{\text{coop}} = \frac{1}{2} \text{ReLU}(1 - \rho(\mathbf{u}, \boldsymbol{\pi})). \quad (17)$$

The Pearson correlation is differentiable everywhere, invariant to an affine change of either map and free of any window size to tune.

Edge supervision. The additive branch is trained to reproduce the gradient that the backbone lost, measured

with the Sobel operator ∇ and weighted toward the places where the clean reference has an edge. Let $\mathbf{w}$ be the clean Sobel magnitude divided by its mean and clipped at ten, $\tilde{\mathbf{w}}$ its square normalized again to unit mean, and $\bar{\mathbf{g}}$ a Sobel magnitude divided by its own maximum. With $\delta = [1.05 \text{EPI}(\mathbf{b}) - \text{EPI}(\hat{\mathbf{x}})]_+$,

$$\begin{aligned} \mathcal{L}_{\text{edge}} = & 10 \left\| \sqrt{\mathbf{w}} \odot (\nabla \hat{\mathbf{x}} - \nabla \mathbf{x}) \right\|_2^2 \\ & + 10 \left\| \sqrt{\tilde{\mathbf{w}}} \odot (\bar{\mathbf{g}}(\hat{\mathbf{x}}) - \bar{\mathbf{g}}(\mathbf{x})) \right\|_2^2 + 45 (3\delta + \delta^2). \end{aligned} \quad (18)$$

The first part is the gradient the correction must restore, the second compares normalized edge magnitude, which is insensitive to overall brightness, and the third refuses an edge preservation index below five percent above the backbone.

Bounds and background protection. The last term collects the constants that keep the change small, smooth and confined. Let a be the mean absolute change over the upper half of the image, e the mean absolute output of the additive edge branch, σ_B the standard deviation over the lower quarter, μ_T the mean over the upper half and $\mathbf{m}$ a soft mask on intensities below the fifteenth percentile of the backbone output,

$$\begin{aligned} \mathcal{L}_{\text{reg}} = & 40[0.002 - a]_+ + 200[a - 0.006]_+^2 + 2 \overline{\|\nabla \boldsymbol{\alpha}\|} \\ & + 30[e - 0.06]_+^2 + 80[\sigma_B(\hat{\mathbf{x}}) - 0.75 \sigma_B(\mathbf{b})]_+ \\ & + 100 \left[\frac{\sigma_B(\hat{\mathbf{x}})}{\sigma_B(\mathbf{b})} - 0.95 \right]_+ + 150 \left[0.99 - \frac{\mu_T(\hat{\mathbf{x}})}{\mu_T(\mathbf{b})} \right]_+ \\ & + 2 \overline{|\hat{\mathbf{x}} - \mathbf{b}| \odot \mathbf{m}}, \end{aligned} \quad (19)$$

where a bar denotes the mean over pixels and the first two parts are clipped at five. The floor stops the wrapper learning to do nothing and the ceiling stops it overshooting fidelity, the third keeps the allocation map smooth so it leaves no Sobel trace of its own, the fourth bounds the edge branch, and the last three hold the background noise and the tissue brightness.

B. Data and optimization schedule

Training uses paired data, a noisy scan and a clean reference averaged over repeated acquisitions. The reference enters the fidelity and clinical terms and supervises the edge branch. Testing uses no clean image at any point, since the six properties read only the noisy input and the backbone output, and the reference only scores the results. The backbone is trained first with the squared error alone and is then frozen. The wrapper follows in two stages. Only the clinical terms are active at first, so the correctors learn a useful direction, and the fidelity term switches on at the midpoint, so they learn how far they may go. Both stages use AdamW with weight decay 10^{-4} on 96×96 crops, in batches of sixteen and of eight for DnCNN and SwinIR, for ten epochs in all, with base learning rates of 2×10^{-4} for the correctors, 5×10^{-4} for the cooperation head and the strength predictor

and 3×10^{-4} for the rule layer. The supplementary file restates the procedure as pseudocode.

The normalization statistics of f are held fixed throughout. DnCNN is the one backbone carrying batch normalization, and left in training mode its buffers move, so the reference against which every clinical change of that backbone is measured would drift while the correction was being fitted against it.

C. How the operating point was chosen

The terms of Equation (1) carry fixed relative weights, the same for every backbone and seed. The clinical term dominates at 6, the property term is 1, the cooperation term is 0.2, and the fidelity term is 0 in the first stage and 1 in the second, which is what the two stage schedule means here, and the edge and regularizer terms are 1, since their internal coefficients already carry their scale. The parser also accepts ten weight flags, for the clinical term, the two contrast measures, fidelity in two forms, the edge term, three per measure preservations and efficiency. Not one of them reaches the loss. Two are printed at start up, which is the only thing that made any of them look active, and the release notes list all ten. Two quantities are selected per backbone on validation, the dead zone d of Equation (14) and the gate of Equation (6). The grid crosses the two gate rules with dead zones of 0.3, 0.6 and 0.9 dB and the tissue-bounded gate with radii of 41 and 61 pixels. It chose the tissue-bounded gate at radius 41 with a dead zone of 0.6 dB for NAFNet and the percentile gate at 0.9 dB for the other three. Everything else is held at one value throughout. Each setting is trained at three seeds and scored on validation. A setting is admissible only if it clears its bounds at its worst seed rather than on average, and among the admissible settings we take the one with the largest lower confidence bound on its weakest clinical measure. When no setting clears the first pair of bounds we relax them one step at a time and record the step that was needed, so a backbone that required a looser condition is visible. Every source of randomness is seeded, up to the nondeterminism of the convolution kernels, and the checkpoints chosen on validation are the ones scored on test.

D. Initialization and the choice of every constant

Every constant appears with its value where it acts and has one of four origins. Fixed structural choices, never tuned, are the sharpness $\kappa = 10$ of Equation (3), the steepness 20 of Equation (6), the window sizes, the quarter resolution of the gain network, the nominal bound ± 0.08 of the edge branch, the thresholds τ_i of Table 2, taken from the clinical literature and unchanged by training to four decimals in every run, and the budgets B and C , tied by the identity of Section IV-E. Set once on the validation split while the method was

built and never revisited are $\alpha_{\max} = 0.15$, the gain bound ± 0.20 , the gate percentile 0.15, the three tolerances of Section IV-G, the half strength background blend and the learning rates. Selected in the grid of Section VI-C, on the validation split only, are the dead zone d and the gate rule. Learned by the objective are β , θ_1 to θ_5 , τ_0 , β_0 and the scale of the edge branch. Every layer uses its default initialization with one deliberate exception, the bias of the brightness channel starting at 0.5 so the raw gain begins at $0.5 \alpha_{\max}$ in flat tissue rather than on a flat region of the loss. Nothing was ever chosen on the test split, and Section VII-D reports how far the results move when the constants of the rule layer are changed.

VII. Tests and Results

A. Setup

Data. PKU37 [9] provides 37 cross sections of 640×640 pixels from a spectral domain scanner, references averaged over 50 frames, split by clean image identity into 27, 5 and 5 to give 1388, 173 and 173 pairs. It is the only training domain. Duke17 [10] provides 17 paired B-scans of 970×520 from another institution, of which we use the 16 the released loader reads, and Duke2013 [11] the 18 volumes of 900×450 of its test split. Both Duke sets come from the same manufacturer and the 2013 evaluation reuses some 2012 material, so we call them two external datasets rather than two independent scanners.

Transfer protocol. Every model is trained once on PKU37 and applied to the Duke sets directly, with nothing fitted, recalibrated or selected on Duke data beyond the per image quantity the gate of Equation (6) computes by the same rule as on PKU37. The original submission instead adapted the wrapper for fifteen further epochs on the other subjects of the target set, leave one subject out. Each of its folds keeps the epoch that scores best on the subject it then reports, so those numbers are selected on the data they describe. We rerun the protocol for the record and give it in the supplementary file as a historical diagnostic.

Measures. Peak signal to noise ratio against the clean reference measures fidelity. Six clinical measures, given as percentage change from backbone to corrected output, are the contrast to noise ratio of tissue against background, the tissue contrast index for dynamic range in tissue, the edge preservation index for agreement with the reference edge map, boundary sharpness for the gradient across layer boundaries, the equivalent number of looks for speckle suppression, and the signal to noise ratio against the noise floor. Each is computed per image then averaged, never pooled, since pooling inflates the result. The 173 pairs come from five clean image identities, so every count below describes repeated acquisitions of five anatomies and not a population.

TABLE 3. PKU37 test set, 173 images, four backbones. The first column is the frozen backbone measured alone. The rest are the change the wrapper makes to it, fidelity in decibels with the spread across three seeds, and the six clinical measures as a percentage, bold where the mean exceeds that spread. Every clinical cell with its own spread is in the supplementary file.

Backbone	Backbone PSNR (dB)	Δ PSNR (dB)	Δ CNR	Δ TCI	Δ EPI	Δ BS	Δ ENL	Δ SNR
NAFNet	30.75	-0.18 ± 0.02	+3.49	+1.91	+1.20	+3.89	+23.74	+11.10
DnCNN	30.68	-0.98 ± 0.21	+11.30	+0.99	+4.14	-1.44	+57.14	+22.95
SwinIR	29.87	-0.17 ± 0.06	+7.33	+11.45	+2.68	+1.82	+28.32	+11.54
KBNet	32.04	-0.90 ± 0.08	+6.93	+5.92	+0.00	+2.58	+25.30	+7.95

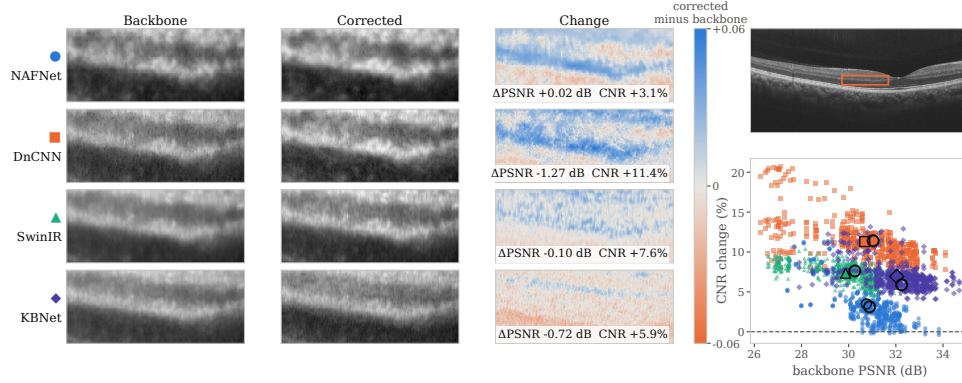


FIGURE 3. The band where the NAFNet correction is largest, 56 by 136 pixels, marked on the thumbnail, one row per backbone. The eight gray cells share one display range set to the values they contain, so nothing is clipped, and the signed change uses a fixed scale of plus or minus 0.06. The scatter places this image, ringed, among all 173 test images at three seeds for four backbones, with the twelve NAFNet points below zero marked. Contrast rises for every backbone and the fidelity it costs differs by more than a decibel between them.

B. In distribution results

Table 3 reports the PKU37 test set, Fig. 4 the corrected scans at full resolution and Fig. 3 the band where the correction is largest. The wrapper raises contrast on every backbone, by 3.5 to 11.3 percent, and the speckle measure by 23.7 to 57.1 percent, for 0.17 dB of fidelity at best and under one decibel at worst. Twenty one of the 24 backbone by measure pairs improve by more than one seed spread, one improves inside it, one is unchanged and one falls, the boundary sharpness of DnCNN. The gain holds image by image and not only in the mean. Contrast rose on all but twelve of the 2076 image and seed combinations, none of the twelve by as much as half a percent, and it is largest where the backbone leaves most to repair.

C. Transfer to other datasets

Table 4 carries the same four models to Duke17 and Duke2013 with nothing refitted. All six measures stay positive on both sets for NAFNet and SwinIR at under 0.35 dB, and 40 of the 48 backbone by measure by set pairs are positive. The eight exceptions are the tissue contrast index and the boundary sharpness of DnCNN and the edge preservation and the signal to noise ratio of KBNet. Contrast rises on every backbone and both sets.

D. Which properties and which constants matter

One property at a time is switched off at test time, by setting its score to the satisfied value and its failure map to zero, and the test set is re evaluated. Only P_5 ,

P_1 and P_2 move any measure by more than a quarter point, costing 3.19, 1.72 and 0.74 points of the speckle gain, while the other three change nothing by more than 0.23. We keep all six, because they are the vocabulary of the rule layer and of the explanation shown to the reader, and one that rarely fires here can fire on another scanner.

We then vary the base level and the two rule weights that carry the allocation, and replace the Lukasiewicz conjunction by the product and by the minimum. The budgets are tied by the identity of Section IV-E and stay fixed. None of the fourteen variations turns a measure negative, contrast staying between 3.5 and 4.5 percent and fidelity between 0.16 and 0.23 dB. Only the base level has a visible effect, lowering it raising contrast and costing fidelity together, while the two weights move the result by under a fifth of a point. The result therefore rests on the shape of the rule layer and not on the particular conjunction.

E. What each component contributes, and how it compares Table 5 removes one component at a time from the selected NAFNet model. Replacing the rule layer by a uniform allocation cuts the contrast gain from 3.7 to 0.8 percent and the tissue contrast index from 2.4 to 0.4 while doubling the fidelity cost, the clearest evidence that the allocation and not the capacity of the correctors produces the clinical gain. Removing the edge branch turns both negative, removing the cooperation map costs about a third of the contrast gain, and removing background smoothing leaves contrast and

TABLE 4. Duke17 and Duke2013, the models of Table 3 applied with no parameter updated, no threshold recalibrated and no statistic of the target images used to select anything. Columns as in Table 3.

Backbone	Backbone PSNR (dB)	Δ PSNR (dB)	Δ CNR	Δ TCI	Δ EPI	Δ BS	Δ ENL	Δ SNR
Duke17								
NAFNet	27.03	-0.17 ± 0.01	+7.51	+4.00	+1.37	+4.15	+21.35	+5.43
DnCNN	26.89	-0.29 ± 0.05	+12.19	-1.47	+3.49	-2.24	+39.42	+13.55
SwinIR	27.06	-0.19 ± 0.05	+8.68	+13.26	+2.69	+1.90	+29.46	+11.60
KBNet	27.59	-0.51 ± 0.05	+5.34	+7.19	-0.50	+1.61	+7.49	-1.81
Duke2013								
NAFNet	27.33	-0.35 ± 0.01	+7.37	+1.41	+0.85	+0.51	+28.80	+9.29
DnCNN	27.21	-0.42 ± 0.05	+12.73	-2.21	+2.99	-5.85	+49.91	+17.59
SwinIR	26.89	-0.19 ± 0.06	+9.26	+11.86	+2.55	+2.30	+30.90	+13.31
KBNet	28.12	-0.59 ± 0.04	+7.16	+4.83	-0.35	+3.06	+8.70	-0.90

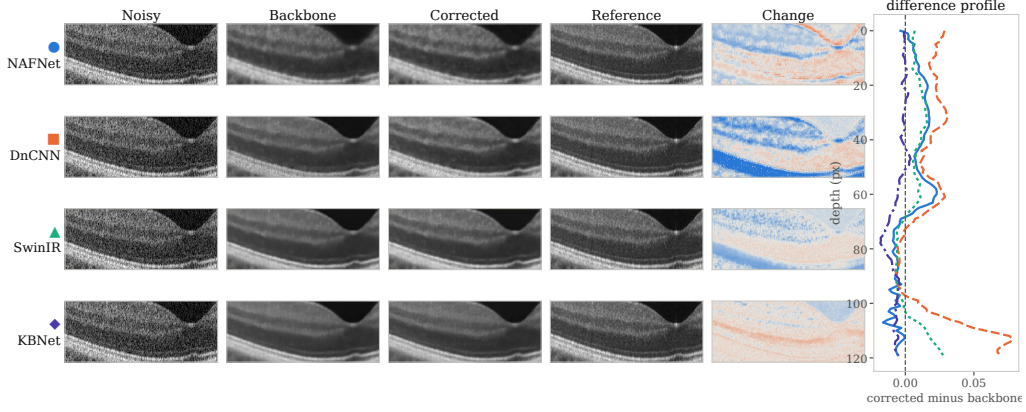


FIGURE 4. The same PKU37 test image at full resolution, a 120 by 300 pixel region spanning the retinal layers, one row per backbone.

Columns are the noisy input, the frozen backbone output, the corrected output, the clean reference, and the signed change the wrapper made, blue where it raised the intensity and orange where it lowered it. The change concentrates on the layer boundaries and the tissue between them and is close to zero in the dark vitreous above. The right panel averages that change across the width, each curve in the color of its row, so the depth at which each backbone is corrected can be read directly.

the two edge measures almost intact while the speckle measure collapses from 23.3 to 5.8 percent.

The lower block spends a similar budget three other ways on the same backbone output. The first is the same architecture, objective and ten epochs with the rule layer replaced by a uniform allocation throughout training rather than only at test time. It was fitted before the gate search, so it carries the percentile gate and not the tissue-bounded gate NAFNet was selected with, which makes it matched in architecture and not in configuration. It reaches a larger contrast gain and a much larger tissue contrast index for 0.42 dB against 0.17 and is worse on four of the six measures, so the two are not ordered. What the full method buys with its fidelity is the account of why each change was made. The other two are classical operators tuned on the validation split. Both reach a tissue contrast index far above anything the wrapper does and both fail elsewhere, lowering edge preservation by about 3.5 percent and the speckle measure by about 36 percent while raising boundary sharpness by about 25 percent by sharpening noise. Adaptive equalization also costs 4.4 dB.

F. Does the cooperation map predict error

Called a confidence map in the original submission, this component was tested against the absolute error of the backbone output over 173 images, and it does not

predict that error. The quantity tested is the demand map $\mathbf{u} = \mathbf{1} - \mathbf{c}$, high where the wrapper is told to act, so a demand map tracking backbone error would give a positive rank correlation. The measured value is -0.258, ordered the wrong way rather than merely uninformative, and removing the pixels it ranks least reliable raises the remaining error to 1.88 times the starting value where an oracle ranking reaches 0.07. The map is nevertheless useful, since replacing it by a constant lowers the contrast gain from +3.71 to +2.51 percent and worsens five of the six measures. Nothing in the objective shows the head the error of the backbone, so we call it a cooperation map and not a confidence.

G. Do the theoretical conditions hold on real data

We perturb the rule truth values and compare the ratio of Equation (11) with the constant of Equation (12), and we measure the tissue gain, the background leakage and the ratio of background standard deviations against the hypotheses of Theorem 2. The bound holds with room to spare, 0.5489 against a largest measured ratio of 0.4679 over 173 images. The constant is exact, so the gap is not slack in the proof. It says the perturbations a real image produces never reach the worst case the constant allows. The allocation map spans 0.556 to 0.793 with mean 0.726, and 0.00 percent of pixels sit at either end

TABLE 5. The selected NAFNet model on the same 173 test images. Above, one component of the wrapper removed at a time. Below, three other ways of spending a similar budget on the same backbone output. Columns as in Table 3 without the backbone column, which is the same throughout.

Variant	Δ PSNR (dB)	Δ CNR	Δ TCI	Δ EPI	Δ BS	Δ ENL	Δ SNR
One component removed							
Full method	-0.17	+3.71	+2.42	+1.21	+4.16	+23.34	+10.74
No rule layer	-0.38	+0.83	+0.38	+0.99	+4.27	+25.45	+10.44
No edge branch	-0.37	-2.71	-6.54	+0.55	+0.65	+20.28	+7.30
No cooperation map	-0.15	+2.51	+1.15	+1.10	+3.87	+23.84	+10.61
No background smoothing	-0.18	+3.03	+6.88	+0.92	+3.99	+5.75	+2.85
Alternatives of similar complexity on the same backbone output							
Uniform allocation, trained	-0.42	+4.11	+11.58	+0.71	+3.83	+16.18	+5.80
Unsharp masking	-0.67	+0.63	+39.45	-3.59	+25.13	-35.31	-20.57
Adaptive equalization	-4.43	+6.48	+53.03	-3.45	+24.62	-38.60	-25.13

of its clamp, direct evidence that the rule layer does work after the repair of Section IV-E.

The margin condition holds on 139 of 173 images and the ratio condition on 94, but both together on only 60, and the theorem needs both. Where a hypothesis fails the theorem predicts nothing, and the measured contrast change there does not differ systematically in sign, so the conditions are sufficient and not necessary. Both leakage terms sit far below the mean absolute change over the image, 0.0067, at -0.0007 in tissue and -0.0017 in background, and the per image sign is the direction the decomposition assumes on 137 and 107 of 173 images.

H. Safety, and whether the correction invents structure

A correction that raises contrast could sharpen a boundary that is not there or erase a faint one that is, and the clean reference lets us measure both. Two measures move against it. An edge appears where the reference has none on 0.48 percent of pixels after correction against 0.45 percent for the backbone, a rise on 127 of 173 images, and the retention of structures in the lowest decile of reference contrast falls from 40.5 to 37.6 percent, on 168. Both are significant under a two sided Wilcoxon signed rank test on the paired per image values, at $p = 1.9 \times 10^{-12}$ and $p = 6.2 \times 10^{-30}$. Neither is zero by construction, since the clamp bounds how far a pixel may move and not how many cross an edge threshold, and Section VIII lists both. What separates the wrapper from a sharpening filter is Table 5, where the classical sharpeners lose edge preservation and the speckle measure while it gains on both.

Three checks come out in its favor. The two layer boundaries barely move, the inner limiting membrane shift falling from 4.73 to 4.38 pixels and the retinal pigment epithelium shift changing by less than a signed rank test can separate from zero. Dark fluid like regions occupy 1.6 percent of the image with a mean absolute change of 0.0022 inside them against 0.0067 overall, the largest change anywhere being 0.0178, so the correction is quieter where a dark region carries the finding. Boundary sharpness is not worsened on 104 of 173 images. Of the three constraints of Section IV-G, energy holds on 173 and bounded change on 173, and only the no sacrifice constraint ever fails, on 173 minus 135. None falls below

two, so the candidate is returned in full everywhere and the layer never acted on this test set.

I. Cost of moving to a new backbone

A new backbone means refitting the whole wrapper, not only its head. The trainable count is 168 630 shared parameters plus a head whose width follows the feature tensor it reads, from 4659 for KNet to 234 273 for DnCNN, where the first convolution is wide. Against a frozen backbone of about seven million that is 2.3 to 5.8 percent, so the shared part is constant and the cost of attaching it follows how wide a tensor the backbone exposes. Every backbone is fitted on the same 1388 pairs for the same ten epochs, so only the wall clock and the seed spread vary. One seed takes 2.0 minutes for NAFNet and 8.3 for SwinIR on a single RTX 5070 Ti, following the cost of one backbone forward pass, and the seed spread of the fidelity cost runs from 0.02 dB for NAFNet to 0.21 for DnCNN. The backbone with the widest spread is the only one whose clinical measure falls.

VIII. Discussion

A. Why fidelity falls, and when that is the right trade

Fidelity falls whenever the wrapper acts, and it has to. A denoiser trained on squared error already sits near a local optimum of that error, so any change that raises contrast or sharpens a boundary moves away from it. What matters is the size of the payment. It is under a decibel, smaller than the difference between two acquisitions of the same eye, and it buys what a grader reads.

Where the gain lands is decided by what each backbone leaves undone rather than by how good it is overall. On boundary sharpness NAFNet starts furthest below the clean reference and gains most, while DnCNN starts above it and is the only backbone that falls. On edge preservation KNet starts highest and is the only one left unchanged. SwinIR is the instructive case, since windowed attention smooths hard inside a window, so it leaves the most room on contrast and pays the least fidelity. A deployment that cannot afford the payment lowers the gain bound $\alpha_{\max}$, though only bypassing the wrapper returns the backbone output exactly.

B. Limitations

The correction is bounded by construction, so a backbone that destroyed a structure cannot be made to recover it. Two safety measures move against it, the rate at which an edge appears where the reference has none and the retention of the weakest structures, both reported in Section VII-H. Neither cost is zero, and the criterion that chose the operating point scores six clinical measures and fidelity and sees neither. We measured what that blindness costs, on validation so that the test split is spent once. The runner up setting for NAFNet keeps more weak structure than the selected one, 43.9 percent against 37.7, and invents fewer edges. A user who values faint detail above contrast should therefore score these two during selection. We did not reselect on this evidence, since the criterion was fixed before the test split was scored. The measures are established proxies, not outcomes, and no clinician read the images, so Section VII-H is quantitative evidence rather than clinical validation. The external sets are small, sixteen and eighteen images, so the transfer results show behavior across scanners rather than a precise estimate of the gain on any one device.

IX. Conclusion

We presented a post-correction layer for OCT denoising, attached to a frozen denoiser rather than replacing it. It states six clinical properties in closed form, turns them into a per-pixel decision through five named rules, and limits how far the change may move the image, so any change traces back to the property that motivated it and the constraint that let it stand. Across four backbone families the same rules, budgets and loss weights raise the clinical measures at a fidelity cost below one decibel, and on three of the four no measure falls at all. The exception is boundary sharpness on DnCNN, which starts above the clean reference, where the method has no room to act. The measures are proxies, and the next step is a reader study asking whether the changes are the ones a grader would want.

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